

# Draft Solutions and Marking Guide

## Ph.D. Macroeconomics II, Questions 2–4

### Question 2: Government spending in the New-Keynesian model

1. Goods clearing implies

$$\hat{c}_t = \frac{\hat{y}_t - s_G \hat{g}_t}{1 - s_G}.$$

Using  $\hat{n}_t = \hat{y}_t$  and  $\widehat{mc}_t = \hat{c}_t + \varphi \hat{n}_t$ ,

$$\widehat{mc}_t = \left( \frac{1}{1 - s_G} + \varphi \right) \hat{y}_t - \frac{s_G}{1 - s_G} \hat{g}_t.$$

Flexible prices imply  $\widehat{mc}_t = 0$ , hence

$$\hat{y}_t^n = \frac{s_G}{1 + \varphi(1 - s_G)} \hat{g}_t.$$

Give up to 2 points for goods clearing/private consumption, 2 points for marginal cost, and 1 point for natural output.

2. Let

$$D \equiv 1 + \varphi(1 - s_G), \quad \psi \equiv \frac{s_G}{D}.$$

Then  $\hat{y}_t^n = \psi \hat{g}_t$ . The natural level of private consumption is

$$\hat{c}_t^n = \frac{\hat{y}_t^n - s_G \hat{g}_t}{1 - s_G} = -\frac{\varphi s_G}{D} \hat{g}_t.$$

The natural real interest rate is

$$r_t^n = E_t \hat{c}_{t+1}^n - \hat{c}_t^n = (1 - \rho_g) \frac{\varphi s_G}{D} \hat{g}_t.$$

The output gap is  $x_t = \hat{y}_t - \hat{y}_t^n$ . Since the private-consumption gap is  $x_t/(1 - s_G)$ , the gap IS equation is

$$x_t = E_t x_{t+1} - (1 - s_G)(\hat{i}_t - E_t \pi_{t+1} - r_t^n).$$

The Phillips curve is

$$\pi_t = \beta E_t \pi_{t+1} + \kappa_y x_t, \quad \kappa_y \equiv \kappa \left( \frac{1}{1 - s_G} + \varphi \right) = \kappa \frac{D}{1 - s_G}.$$

Give 2 points for defining the gap and the Phillips curve, 2 points for the correctly scaled IS equation, and 1 point for  $r_t^n$ .

3. Guess  $x_t = a_x \hat{g}_t$  and  $\pi_t = a_\pi \hat{g}_t$ . Since  $E_t \hat{g}_{t+1} = \rho_g \hat{g}_t$ ,

$$(1 - \rho_g)a_x + (1 - s_G)(\phi_\pi - \rho_g)a_\pi = (1 - s_G)(1 - \rho_g)\frac{\varphi s_G}{D},$$

and

$$(1 - \beta \rho_g)a_\pi = \kappa_y a_x.$$

Thus

$$a_x = \frac{(1 - s_G)(1 - \rho_g)\frac{\varphi s_G}{D}}{(1 - \rho_g) + (1 - s_G)(\phi_\pi - \rho_g)\frac{\kappa_y}{1 - \beta \rho_g}},$$

and

$$a_\pi = \frac{\kappa_y}{1 - \beta \rho_g} a_x.$$

Both coefficients are positive under the stated assumptions. A higher  $\phi_\pi$  raises the denominator and therefore lowers  $a_x$  and  $a_\pi$ . Give 3 points for the algebraic solution and 2 points for signs/comparative statics.

4. Under flexible prices,  $x_t = 0$ ,  $\pi_t = 0$ , and output rises by

$$\hat{y}_t^n = \frac{s_G}{D} \hat{g}_t.$$

Private consumption falls because government purchases absorb resources and create a negative wealth effect:

$$\hat{c}_t^n = -\frac{\varphi s_G}{D} \hat{g}_t < 0.$$

Under sticky prices and the Taylor rule, the government-spending shock raises the natural real rate. Since the nominal-rate rule does not perfectly replicate the natural real rate, the

real rate is too low relative to  $r_t^n$ , so  $x_t > 0$  and  $\pi_t > 0$ . Output rises by more than under flexible prices:

$$\hat{y}_t = \hat{y}_t^n + x_t.$$

Private consumption may fall by less than under flexible prices, and can in principle rise if the positive output gap is large enough, since

$$\hat{c}_t = \hat{c}_t^n + \frac{x_t}{1 - s_G}.$$

Optimal monetary policy stabilizes the output gap and inflation, implementing  $x_t = 0$  and  $\pi_t = 0$  by setting the real rate equal to the natural real rate. Give 2 points for flexible prices, 2 points for sticky-price mechanism, and 1 point for optimal policy.

### Question 3: Reservation wages and wage dispersion

1. The Bellman equations are

$$\begin{aligned} rW(w) &= w + \sigma[U - W(w)], \\ rU &= b + \lambda_u \int \max\{W(w) - U, 0\} dF(w). \end{aligned}$$

Since  $W(w)$  is increasing in  $w$ , there is a cutoff  $w_R$  such that the worker accepts iff  $w \geq w_R$ , with  $W(w_R) = U$ . Give 2 points for Bellman equations and 2 points for cutoff logic.

2. From the employed Bellman equation,

$$W(w) = \frac{w + \sigma U}{r + \sigma}.$$

Using  $W(w_R) = U$  gives  $rU = w_R$ . Hence

$$W(w) - U = \frac{w - w_R}{r + \sigma}.$$

Substitute into the unemployment Bellman equation:

$$w_R = b + \lambda_u \int_{w \geq w_R} \frac{w - w_R}{r + \sigma} dF(w),$$

or

$$w_R - b = \frac{\lambda_u}{r + \sigma} \int_{w \geq w_R} (w - w_R) dF(w).$$

The reservation wage increases with  $b$  and  $\lambda_u$ , and decreases with  $\sigma$ . Intuitively, higher benefits or more frequent offers make waiting more valuable; faster separations make accepted jobs less valuable. Give 3 points for derivation and 2 points for comparative statics.

3. Since

$$\int_{w \geq w_R} (w - w_R) dF(w) = Ew - w_R - \int_{\underline{w}}^{w_R} (w - w_R) dF(w),$$

and

$$-\int_{\underline{w}}^{w_R} (w - w_R) dF(w) = \int_{\underline{w}}^{w_R} (w_R - w) dF(w) = \int_{\underline{w}}^{w_R} F(w) dw,$$

we obtain

$$w_R - b = \frac{\lambda_u}{r + \sigma} \left[ Ew - w_R + \int_{\underline{w}}^{w_R} F(w) dw \right].$$

For a given reservation wage, a mean-preserving spread increases the option value of search because the worker can reject sufficiently bad offers while preserving the upside from very good offers. Therefore the right-hand side of the reservation-wage equation rises at the old reservation wage, and the equilibrium reservation wage increases. Give 2 points for the transformation and 2 points for the economic interpretation.

4. A higher reservation wage lowers the acceptance probability  $1 - F(w_R)$  and therefore increases expected unemployment duration. Accepted wages are drawn from a higher part of the offer distribution, so average accepted wages and match quality rise. Give 1 point for unemployment duration and 1 point for accepted wages.

## Question 4: Idiosyncratic risk in the Aiyagari model

1. The household problem is

$$V(a, \epsilon) = \max_{c, a'} \{u(c) + \beta E[V(a', \epsilon') | \epsilon]\}$$

subject to

$$c + a' = w\epsilon + (1 + r)a, \quad a' \geq 0, \quad c \geq 0.$$

The firm solves

$$\max_{K, L} K^\alpha L^{1-\alpha} - (r + \delta)K - wL.$$

The first-order conditions are

$$r + \delta = \alpha K^{\alpha-1} L^{1-\alpha}, \quad w = (1 - \alpha) K^{\alpha} L^{-\alpha}.$$

With mean labor efficiency equal to one,  $L = 1$  in stationary equilibrium. Give 2 points for household problem and 2 points for firm FOCs.

2. A stationary recursive competitive equilibrium consists of prices  $(r, w)$ , policy functions  $c(a, \epsilon)$  and  $a'(a, \epsilon)$ , firm choices  $(K, L)$ , and an invariant distribution  $\mu(a, \epsilon)$  such that: households solve their recursive problem; firms optimize;  $\mu$  is invariant under the household policy and the Markov process for  $\epsilon$ ; labor clears with  $L = \int \epsilon d\mu = 1$ ; asset/capital markets clear with

$$K = \int a d\mu;$$

and the goods market clears,

$$\int c(a, \epsilon) d\mu + \delta K = K^{\alpha} L^{1-\alpha}.$$

Give up to 4 points for these equilibrium components.

3. The Aiyagari diagram plots aggregate asset supply  $A^s(r)$  from household saving behavior and capital demand  $K^d(r)$  from firm optimality. Capital demand is downward sloping:

$$K^d(r) = \left( \frac{\alpha}{r + \delta} \right)^{1/(1-\alpha)}$$

when  $L = 1$ . Aggregate asset supply is increasing in  $r$ : a higher return makes saving more attractive. With uninsured idiosyncratic risk and prudence, households have a precautionary-saving motive. As  $r$  approaches  $1/\beta - 1$  from below, desired saving becomes very large. A stationary equilibrium therefore requires

$$r < 1/\beta - 1.$$

Give 1 point for demand, 1 point for supply/diagram, and 1 point for the interest-rate bound.

4. More idiosyncratic income risk strengthens precautionary saving. For any given  $r$ , households want to hold more assets, so  $A^s(r)$  shifts to the right. In equilibrium, the interest rate falls and the capital stock rises. The higher capital stock raises wages and lowers the marginal product

of capital. In a representative-agent or complete-markets economy with the same mean labor endowment, purely idiosyncratic risk is insured or diversified away, so this precautionary-saving channel is absent. Give 2 points for the shift and equilibrium effects, and 2 points for the complete-markets contrast.